PROJECT 10 / SCIENTIFIC WEB APPLICATION

# Nuclear Astrophysics Lab

## Verification & data provenance

Version1.0.0 | 2026-10-15 | Independent review package

### Ten calculations. Explicit assumptions. Traceable data.

The source, production build, offline HTML, numerical datasets, tests and documentation are included. The application has not been published. This report separates successful executed numerical checks from browser checks that could not be executed in this environment.

| | |
|---|---|
| **Implementation** | 10 scientific calculators;15 navigable views; English/Arabic controls; dark/light themes; PDF/CSV/PNG exporters. |
| **Curated database** | 5 reactions; 8 data/model records. Shared experimental input is identified; these are not eight independent evaluations. |
| **Executed tests** | 128/128 passed; 0 failed. This includes89 physics/unit/numerical checks and39 data/report/render/scenario checks. |
| **Reference cases** | 10/10 pass independent analytic or rounded published reference criteria. |
| **Data validation** | 382 checks passed, including every original NACRE row and all six pinned REACLIB coefficient sets. |
| **Browser status** | 28 desktop/mobile checks skipped;0 browser checks passed. Chromium was absent. Server rendering does not replace browser interaction or visual QA. |
| **Build checks** | npm ci, npm test, npm run typecheck, npm run validate:data and npm run build completed. Normal production and self-contained HTML builds generated. |
| **Main review limit** | No monoenergetic total experimental sigma(E) grid is bundled. Thermal integration uses explicit mathematical models or user-provided CSV; missing tails and covariance are not invented. |

This application provides educational and research-oriented nuclear astrophysics calculations under explicitly stated assumptions. It is not a complete stellar evolution or nucleosynthesis simulation.

# Calculation reference checks

Expected values are independent analytic limits or the stated published rounded formula. Relative differences are numerical comparisons, not experimental uncertainty.

## 01 Finite-range PDF normalization

| | |
|---|---|
| **Question** | How is relative kinetic energy distributed in a classical thermal plasma? |
| **Inputs** | T=15 MK; mu=.5 u; E=0 to50kT |
| **Equation** | f(E) = 2/sqrt(pi) * sqrt(E)/(kT)^(3/2) * exp(-E/kT) |
| **Reference / source** | Analytic gamma-distribution integral, integral f(E)dE=1; no engine-generated expected value. |
| **Reference value** | 1.00000000000e+00 1 |
| **Application value** | 1.00000000000e+00 1 |
| **Difference** | Signed: 0.00000e+00 1; relative absolute: 0.00000e+00% |
| **Tests / result** | PASS \| relative tolerance 1.00e-08 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Classical nondegenerate equilibrium; finite energy normalization. |

## 02 S(E)

| | |
|---|---|
| **Question** | What S(E) corresponds to an energy-dependent charged-particle cross section? |
| **Inputs** | E=100keV; sigma=1e-6barn; Z1=Z2=1; mu=1u |
| **Equation** | S(E) = sigma(E) E exp(2 pi eta); 2 pi eta = 2 pi alpha Z1 Z2 sqrt(mu c^2 / (2 E)) |
| **Reference / source** | Rauscher 2010 eq4,5; independently evaluated Python numeric reference using CODATA2022. |
| **Reference value** | 2.28535359759e-03 keV barn |
| **Application value** | 2.28535359759e-03 keV barn |
| **Difference** | Signed: 2.60209e-18 keV barn; relative absolute: 1.13859e-13% |
| **Tests / result** | PASS \| relative tolerance 1.00e-10 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Charged particles only. Point extraction; sigma-only error optional. |

# Calculation reference checks / continued

Expected values are independent analytic limits or the stated published rounded formula. Relative differences are numerical comparisons, not experimental uncertainty.

## 03 Gamow E0

| | |
|---|---|
| **Question** | Where does the nonresonant Coulomb-Maxwell kernel reach its maximum? |
| **Inputs** | T9=.015; Z1=Z2=1; mu=.5u |
| **Equation** | EG = 2 mu c^2 (pi alpha Z1 Z2)^2; E0 = [EG (kT)^2 / 4]^(1/3) |
| **Reference / source** | Rauscher 2010 eq8, published rounded prefactor0.12204. |
| **Reference value** | 5.89141622486e+00 keV |
| **Application value** | 5.89150530294e+00 keV |
| **Difference** | Signed: 8.90781e-05 keV; relative absolute: 1.51200e-03% |
| **Tests / result** | PASS \| relative tolerance 6.00e-05 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Nonresonant, unscreened, slowly varying S. Not general resonant effective energy. |

## 04 Full Gaussian1/e width

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|---|---|
| **Question** | What is the approximate full 1/e width around the Gamow peak? |
| **Inputs** | T9=.015; Z1=Z2=1; mu=.5u |
| **Equation** | Delta = 4 sqrt(E0 kT / 3); interval = [max(0,E0-Delta/2), E0+Delta/2] |
| **Reference / source** | Rauscher 2010 eq9, published rounded prefactor0.23682. |
| **Reference value** | 6.37268952081e+00 keV |
| **Application value** | 6.37300962567e+00 keV |
| **Difference** | Signed: 3.20105e-04 keV; relative absolute: 5.02307e-03% |
| **Tests / result** | PASS \| relative tolerance 6.00e-05 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Full1/e width, not FWHM or hard energy bounds. |

# Calculation reference checks / continued

Expected values are independent analytic limits or the stated published rounded formula. Relative differences are numerical comparisons, not experimental uncertainty.

## 05 Covered-interval <sigma v>

| | |
|---|---|
| **Question** | What is the cross-section-weighted thermal mean of relative speed? |
| **Inputs** | T=1e8K; mu=1u; sigma(E)=1barn; E=1e-8 to1e4keV |
| **Equation** | <sigma v> = sqrt(8/(pi mu)) (kT)^(-3/2) integral sigma(E) E exp(-E/kT) dE |
| **Reference / source** | Independent analytic sigma*sqrt(8kT/pi mu); endpoints negligibly truncate the full integral. |
| **Reference value** | 1.45508093160e-16 cm3/s |
| **Application value** | 1.45508093160e-16 cm3/s |
| **Difference** | Signed: -4.93038e-31 cm3/s; relative absolute: 3.38839e-13% |
| **Tests / result** | PASS \| relative tolerance 1.00e-08 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Declared benchmark, not measured nuclear data; finite data never implies full coverage. |

## 06 Identical-reactant event rate

| | |
|---|---|
| **Question** | How many two-body reactions occur per volume and time? |
| **Inputs** | n=2e10cm^-3; k=1e-20cm3/s |
| **Equation** | r = n1 n2 <sigma v> / (1 + delta12) |
| **Reference / source** | Unordered-pair counting r=n^2 k/2, analytic arithmetic. |
| **Reference value** | 2.00000000000e+00 reactions/cm3/s |
| **Application value** | 2.00000000000e+00 reactions/cm3/s |
| **Difference** | Signed: 0.00000e+00 reactions/cm3/s; relative absolute: 0.00000e+00% |
| **Tests / result** | PASS \| relative tolerance 1.00e-12 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Each event consumes two reactants; homogeneous binary model. |

# Calculation reference checks / continued

Expected values are independent analytic limits or the stated published rounded formula. Relative differences are numerical comparisons, not experimental uncertainty.

## 07 Fixed-reservoir depletion scale

| | |
|---|---|
| **Question** | What is the local depletion scale under fixed conditions? |
| **Inputs** | n_partner=1e20cm^-3; k=1e-22cm3/s |
| **Equation** | tau = 1 / (n_partner <sigma v>); a+a: n(t)=n0/(1+t/tau0) |
| **Reference / source** | Analytic solution dn_a/dt=-n_b k n_a; tau=1/(n_b k). |
| **Reference value** | 1.00000000000e+02 s |
| **Application value** | 1.00000000000e+02 s |
| **Difference** | Signed: 2.27374e-13 s; relative absolute: 2.27374e-13% |
| **Tests / result** | PASS \| relative tolerance 1.00e-12 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Fixed temperature and partner reservoir. Not radioactive half-life. |

## 08 Isolated narrow-resonance molar contribution

| | |
|---|---|
| **Question** | What is the isolated on-resonance thermal contribution? |
| **Inputs** | T9=.3; mu=14/15u; Er=259.4keV; omegaGamma=.0129eV; Gamma=.96keV |
| **Equation** | <sigma v>r = (2 pi/(mu kT))^(3/2) hbar^2 (omega gamma) exp(-Er/kT) |
| **Reference / source** | Longland2010 eq10 rounded1.5399e11,11.605; Imbriani2005 measured CM parameters. |
| **Reference value** | 5.88079476360e-01 cm3/mol/s |
| **Application value** | 5.88178721093e-01 cm3/mol/s |
| **Difference** | Signed: 9.92447e-05 cm3/mol/s; relative absolute: 1.68761e-02% |
| **Tests / result** | PASS \| relative tolerance 2.00e-04 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | On-resonance contribution only; finite-width tails/partial-width variation not included. Rounded-reference difference expected. |

# Calculation reference checks / continued

Expected values are independent analytic limits or the stated published rounded formula. Relative differences are numerical comparisons, not experimental uncertainty.

## 09 Local logarithmic temperature slope

| | |
|---|---|
| **Question** | How strongly does the reaction rate respond to temperature? |
| **Inputs** | rate=7*T9^2.5; nonuniform grid [.01,.017,.03,.1,.8,1] |
| **Equation** | nu(T) = d ln(rate) / d ln(T) |
| **Reference / source** | Exact analytic derivative dln(7T^2.5)/dlnT=2.5. |
| **Reference value** | 2.50000000000e+00 1 |
| **Application value** | 2.50000000000e+00 1 |
| **Difference** | Signed: -3.55271e-15 1; relative absolute: 1.42109e-13% |
| **Tests / result** | PASS \| relative tolerance 1.00e-10 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | One-sided boundary derivatives flagged; no extrapolation. |

## 10 Specific local energy generation

| | |
|---|---|
| **Question** | How much energy is deposited locally by the specified events? |
| **Inputs** | r=1e10events/cm3/s; Edep=1MeV; rho=100g/cm3 |
| **Equation** | qV = r Edep; epsilon = qV/rho; 1 erg/g/s = 1e-4 W/kg |
| **Reference / source** | Exact eV-to-joule SI conversion; epsilon=r Edep/rho. 1erg/g/s=1e-4W/kg. |
| **Reference value** | 1.60217663400e+02 erg/g/s |
| **Application value** | 1.60217663400e+02 erg/g/s |
| **Difference** | Signed: 0.00000e+00 erg/g/s; relative absolute: 0.00000e+00% |
| **Tests / result** | PASS \| relative tolerance 1.00e-12 \| tests/physics.test.ts; executed reference generator scripts/validation-report.ts |
| **Limits** | Energy deposited locally is explicit. Neutrino/nonlocal losses excluded. |

# Nuclear data & source versions

The database JSON contains source IDs, reaction identity/products, types, ranges, values, uncertainty and missing-data status. Only small factual extracts and verified selected fit coefficients are bundled. Nuclear databases are not relabeled MIT or CC0.

## NACRE II · 2H(p,gamma)3He

Xu et al. 2013; official table, accessed 2026-10-15

Evaluated total rate, 60 tabulated temperatures

Uncertainty: Published low/high evaluation bounds; no covariance. Not assigned a confidence level here.

https://www.astro.ulb.ac.be/nacreii/data/2hpg3he.dat

## NACRE II · 3He(3He,2p)4He

Xu et al. 2013; official table, accessed 2026-10-15

Evaluated total rate, 58 tabulated temperatures

Uncertainty: Published low/high evaluation bounds; no covariance.

https://www.astro.ulb.ac.be/nacreii/data/3he3he2p4he.dat

## JINA REACLIB · ls09

2025-03-30 snapshot via pynucastro; pinned 0c22729e39bc14b0510a03f53faf92526b0a2bd6

Two additive fit sets; app interval T9=0.01–1 is a policy, source validity not certified

Uncertainty: Fit covariance and nuclear-data uncertainty unavailable.

https://reaclib.jinaweb.org/c12%28p%2Cg%29n13/ls09/

## JINA REACLIB · im05

2025-03-30 snapshot via pynucastro; pinned 0c22729e39bc14b0510a03f53faf92526b0a2bd6

Four additive fit sets; original fit T9<2; app T9=0.01–1

Uncertainty: Fit covariance and nuclear-data uncertainty unavailable.

https://reaclib.jinaweb.org/n14%28p%2Cg%29o15/im05/

## Imbriani et al. · total rate Eq. 15

arXiv:nucl-ex/0509005v1 (2005)

Published analytical total-rate fit; T9<2; app T9=0.01–1

Uncertainty: Paper states fit precision 0.5% (representation error, not total nuclear uncertainty). No combined uncertainty assigned.

https://arxiv.org/abs/nucl-ex/0509005

## LUNA · 6.79 MeV capture transition

Imbriani et al. 2005, Table 2; 11 points below resonance

Measured PARTIAL-channel S factors; Ecm=118.9–248.6 keV. Not total capture.

Uncertainty: Table error bars retained as quoted; normalization/systematic covariance unavailable.

https://arxiv.org/abs/nucl-ex/0509005

## LUNA · 259.4 keV resonance

Imbriani et al. 2005 §3.3, CM energy and width

Measured isolated resonance; app T9=0.2–1; this interval is a conservative policy, not a published guarantee

Uncertainty: Er ±0.3 keV, Gamma ±0.05 keV; strength ±0.4 stat ±0.8 systematic meV, kept separate.

https://arxiv.org/abs/nucl-ex/0509005

## KADoNiS · 56Fe(n,gamma)57Fe

Historical v0.0, last review 1999; CFM91/92; accessed 2026-10-15

11 published MACS and molar-rate values; kT=5–100 keV; SEF=1.00 in this table

Uncertainty: MACS30=11.7±0.5 mb. No per-temperature covariance or rate confidence interval assigned.

https://exp-astro.de/kadonis/selementquery.php?isotope=Fe56

# Uncertainty & numerical interpretation

## Nuclear data

NACRE low/high evaluation bounds are preserved without assigning a new confidence level. JINA fit covariance is unavailable. LUNA quoted partial-channel errors and resonance strength components remain separately identified.

## Thermal / density inputs

Temperature sensitivity is calculated, but no missing temperature or density uncertainty is fabricated. Mass-number defaults are explicit approximations. No full propagated uncertainty is available without input errors and covariance.

## Resonance record

Er=259.4+-0.3keV and Gamma=0.96+-0.05keV are CM. Strength=12.9+-0.4stat+-0.8sysmeV. Statistical and systematic rate effects are separate; width is primarily a model-validity diagnostic. User-modified parameters do not inherit measurement errors.

## Numerical integration

Embedded Gauss7-Kronrod15 quadrature in log energy, source breakpoints and a tighter second pass are required. Nonconvergence withholds the result. Numerical error excludes missing tails, unresolved structures, interpolation and model error.

## Energy accounting

Energy deposited locally is a required explicit input. Escaping neutrinos and nonlocal photon energy are excluded. If available energy is supplied, Edep cannot exceed it. The app never calculates a new mass-derived Q value.

## Correlation

A separate covariance utility requires an explicit symmetric positive-semidefinite matrix. No universal independent-input assumption is made.

## Report consistency

A shared immutable calculation snapshot supplies dashboard results and exports. Reports retain more digits than rounded dashboard cards. Sample PDF/CSV are generated from the same snapshot. The included sample PNG is a standard scientific plot generated by Python, not a browser screenshot or a claim that the browser PNG download was tested.

# Remaining verification & publication

| | |
|---|---|
| **Not executed** | 28 Playwright cases:14 scenarios on desktop/mobile, covering selector, temperature, density, units, charts, downloads, languages, themes, invalid inputs and15-page navigation. Chromium is not installed here. |
| **How to execute** | npm ci; npx playwright install chromium; npm run build; npm run test:e2e. The CI workflow installs Chromium with dependencies and runs the same tests. |
| **No public release** | No GitHub repository creation, source push, Pages enablement or deployment was performed. The included Pages workflow is manual-only and must be dispatched only after explicit approval. |
| **Offline review** | Open standalone.html. Standard dist/index.html should be served over HTTP under /NuclearAstrophysics-Lab/. Browser behavior remains subject to the unexecuted checks above. |
| **Scope exclusions** | No full reaction network, reverse rates, transport, equation of state, stellar equilibrium, composition prediction or stellar evolution. |
| **Complete documentation** | README.md plus docs/SCIENTIFIC_METHODS.md, STELLAR_REACTION_RATES.md, GAMOW_MODELS.md, NUCLEAR_DATA_SOURCES.md, NUMERICAL_INTEGRATION.md, SCIENTIFIC_VALIDATION.md, UNCERTAINTY_POLICY.md, CALCULATION_UNIQUENESS.md, TEST_REPORT.md and KNOWN_LIMITATIONS.md. |